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**RESEARCH FOCUS**

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**Machine Learning:** LLM Agent, World Model, Prompt Learning, Model Pruning, Adversarial ML.**Optimization:** Black-box Optimization, Zeroth-order Optimization, Bi-level Optimization.

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**SCHOOL**

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|-------------------------------------------------------------------------|-------------------------------------------------------------------|------------------------|
| <b>Ph.D.</b> in CSE, Michigan State University (MSU)                    | Advisor: <a href="#">Sijia Liu</a>                                | Jan. 2021 – May. 2025  |
| <b>Ph.D. student</b> in CSE, Tsinghua/MSU                               | Advisor: <a href="#">Zhichao Cao</a> , <a href="#">Yunhao Liu</a> | Aug. 2018 – Dec. 2020  |
| <b>B.S.</b> in Automation, Tsinghua University                          | Advisor: <a href="#">Hong Wang</a>                                | Aug. 2014 – July. 2018 |
| <b>Exchange</b> in CSE, École Polytechnique Fédérale de Lausanne (EPFL) |                                                                   | Aug. 2016 – Feb. 2017  |

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**WORK**

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|-------------------------------------------------|-------------------------------------|------------------------|
| <b>Research Intern</b> in Cisco Research        | Advisor: <a href="#">Gaowen Liu</a> | Feb. 2023 – Jun. 2024  |
| <b>Research Intern</b> in MIT-IBM Watson AI Lab | Advisor: <a href="#">Quanfu Fan</a> | May. 2021 – Aug. 2021  |
| <b>Research Intern</b> in DiDi AI Lab           | Advisor: <a href="#">Yashu Liu</a>  | Nov. 2017 – Feb. 2018  |
| <b>Research Intern</b> in HKUST                 | Advisor: <a href="#">Pan Hui</a>    | June. 2017 – Sep. 2017 |

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**SELECTED PUBLICATION**

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[Google Scholar](#) Citations: 1557

- [1] **Y. Yao\***, Y. Chen\*, Y. Zhang, B. Shen, G. Liu, S. Liu, [Safety Mirage: How Spurious Correlations Undermine VLM Safety Fine-Tuning and Can Be Mitigated by Machine Unlearning](#), *ICLR 2026*.
- [2] K. Chen, Z. Lin, Z. Xu, Y. Shen, **Y. Yao**, J. Rimchala, J. Zhang, L. Huang, [R2I-Bench: Benchmarking Reasoning-Driven Text-to-Image Generation](#), *ACL 2025*
- [3] **Y. Yao\***, J. Liu\*, Y. Gong\*, X. Liu, Y. Wang, X. Lin, S. Liu, [Can Adversarial Examples Be Parsed to Reveal Victim Model Information?](#), *WACV 2025*.
- [4] **Y. Yao\***, Z. Pan\*, G. Liu, B. Shen, H. Zhao, R. Kompella, S. Liu, [From Trojan Horses to Castle Walls: Unveiling Bilateral Backdoor Effects in Diffusion Models](#), *NeurIPS 2024*.
- [5] **Y. Yao**, G. Xiao, V. Asnani, Y. Gong, J. Liu, X. Lin, X. Liu and S. Liu, [Reverse Engineering of Deceptions on Machine- and Human-Centric Attacks](#), *Foundations and Trends in Privacy and Security 2024*.
- [6] S. Pal, **Y. Yao**, R. Wang, B. Shen, S. Liu, [Backdoor Secrets Unveiled: Identifying Backdoor Data with Optimized Scaled Prediction Consistency](#), *ICLR 2024*.
- [7] J. Jia\*, J. Liu\*, P. Ram, **Y. Yao**, G. Liu, Y. Liu, P. Sharma, S. Liu, [Model Sparsity Can Simplify Machine Unlearning](#), *NeurIPS 2023 **Spotlight***.
- [8] A. Chen, **Y. Yao**, P. Chen, Y. Zhang, S. Liu, [Understanding and Improving Visual Prompting: A Label-Mapping Perspective](#), *CVPR 2023*.
- [9] **Y. Yao\***, Y. Zhang\*, P. Ram, P. Zhao, T. Chen, M. Hong, Y. Wang, S. Liu, [Advancing Model Pruning via Bi-level Optimization](#), *NeurIPS 2022*.
- [10] A. Chen\*, P. Lorenz\*, **Y. Yao**, P. Chen, S. Liu, [Visual Prompting for Adversarial Robustness](#), *ICASSP 2022*.
- [11] **Y. Yao\***, Y. Gong\*, Y. Li, Y. Zhang, X. Liu, X. Lin, S. Liu, [Reverse Engineering of Imperceptible Adversarial Image Perturbations](#), *ICLR 2022*.
- [12] Y. Zhang, **Y. Yao**, J. Jia, J. Yi, M. Hong, S. Chang, S. Liu, [How to Robustify Black-Box ML Models? A Zeroth-Order Optimization Perspective](#), *ICLR 2022 **Spotlight***.
- [13] **Y. Yao**, Z. Ma, Z. Cao, [LoSee: Long-Range Shared Bike Communication System Based On LoRaWAN Protocol](#), *EWSN 2019 **Best Poster***.

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**TALKS**

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- 11/2024: **Large Vision Language Model**. Invited Talk at Wayne State University.
- 10/2023: **Machine Unlearning via Model Sparsification**. Invited Talk at Cisco.
- 06/2023: **Reverse Engineering of Deceptions**. CVPR 2023.
- 06/2022: **How to Robustify Black-box Models?** Invited Talk at UIUC.

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**TEACH**

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- **CSE 480: Database Systems** of MSU, Fall 2020, Fall 2023, Spring 2024, Fall 2024.
- **CSE 801: Big Data Analysis** of MSU, Spring 2023.
- **CSE 232: C++** and **CSE 231 & CSE 102: Python** of MSU, Spring & Fall 2021.
- **CSE 477: Web Applications** of MSU, Spring 2021.
- **CSE 891: Artificial Intelligence and Internet of Things (AIOT)** of MSU, Spring 2020.

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**HONOR**

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- **Travel Grant** at NeurIPS 2022, CVPR 2023, NeurIPS 2024.
- **Best Poster Award** at EWSN 2019.
- **Outstanding Undergraduate** of Automation Department, Tsinghua University, 2018.
- **Excellent Academic Scholarship** of Tsinghua University, 2015, 2016, 2017.

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**SERVICE**

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- **Chair** of [AdvML Frontiers](#) at ICML'22, ICML'23, and NeurIPS'24.
- **Reviewers** of NeurIPS, ICLR, ICML, ACL, CVPR, ACMMM, ICASSP; TPAMI.

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**PATENT**

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- Model Assembly with Knowledge Distillation, US18243259
- Domain Adaptation through Model Pruning, US18598148
- Modality-Agnostic Diffusion Prompting, Pending
- Semantic Segmentation using LLM Supervision, Pending
- Bi-Directional LoRA for Machine Unlearning and Retaining Information, Pending
- Generation Model Pruning for Fairness, Pending
- Mixed-Precision Quantization for Machine Unlearning Through Contrastive Learning, Pending
- Optimizing Automated Machine Learning Model Continual Learning Systems, Pending
- Continual Rule Learning for Large Language Models to Reason with Tools, Pending
- Automatic Schema Discovery with Auto-Ingestion and Retrieval Planning, Pending

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**MENTORSHIP**

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- Zhuoshi Pan, B.S./M.S. in Automation, Tsinghua University. Poisoning Diffusion, NeurIPS 2024.
- Brian Zhang, Northville High School, MI. Visual Prompt Ensemble, ICASSP 2024.
- Aochuan Chen, B.S. in Automation, Tsinghua University. Robust Visual Prompt, ICASSP 2023.

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**FUNDING**

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- Cisco Research Award for “Towards Lifelong LMM Agents in Embodied AI” – Research Gift of \$75,000 (Co-lead in proposal development)